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A palaeomagnetic study of some Pleistocene sediments in northern Canada and its bearing on the secular variation of the geomagnetic field

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Summary. Palaeomagnetic data are reported from two 6-m sequences of sediments exposed in north-western Canada near latitude 68°N. The sequences span the Old Crow Tephra which has recently been dated at 86 000± 8000 yr BP. During laboratory treatment a single sample suggests the presence of an underlying reversed magnetization, but no other trace of the Blake Event was found. However, the secular variation pattern observed is of some significance. It consists of a spatially isotropic dispersion due to typical non-dipole field behaviour on which are superimposed linear perturbations due to stationary sources at the core–mantle boundary (CMB). A simple magnetostatic model indicates that a radial dipole just below the CMB need only achieve a maximum amplitude of about 8 per cent that of the geocentric dipole to explain the observations. An earlier study at latitude 50°N revealed evidence of similar behaviour although the sediments involved were much younger than those of the present study. The interesting possibility of 'standing' secular variation sources with lifetimes up to 10^5 yr is therefore raised.

Key words: secular variation, magnetostratigraphy, Quaternary, Blake Event, core–mantle boundary

A Introduction

In recent years there has been a vigorous renewal of interest in the long-term behaviour of the geomagnetic field (Merrill & McElhinny 1983; Creer, Tucholka & Barton 1983). Since a major part of the spectrum involved is associated with time scales in excess of centuries there is a need for appropriate palaeomagnetic data, and one important approach therefore has been the study of long sequences of sedimentary strata. Depending on the time scale in question, the phenomena of interest may be polarity reversals, geomagnetic excursions, or 'normal' secular variation. Also of considerable global interest is the time-averaged morphology of the field in terms of the relative importance of the dipole and higher-order terms in

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the spherical harmonic representation, and the relative contributions of drifting and standing sources to the observed secular variation.

In this paper we report a palaeomagnetic investigation of two 6-m sections of sediments exposed in the banks of the Old Crow River in northwestern Canada. These sediments were deposited in a large meltwater lake and are now preserved in what is called the Old Crow Basin. This is one of three major sedimentary basins in the northern Yukon, which itself is part of a much broader region stretching eastwards from the Bering Straits to the Richardson Mountains, and southwards from the Beaufort Sea to Whitehorse. Even during periods of advanced glaciation this region – often referred to as Beringia – remained unglaciated and therefore served as a refuge for plants and animals, possibly including the first human beings in North America. Fuller accounts of the archeological, geological, and geomorphological details can be found in papers by Hughes (1969, 1972) and Morlan (1980). Because of its high latitude (67°51′N), the site is of special interest to global secular variation models, and since the age of the sediments is about 100 000 yr BP, the results can be expected to have some bearing on the so-called Blake Event.

B Field and laboratory details

Samples were collected from the upper half of unit 3 of Morlan (1980), which at the locality in question (67°51′N, 139°48′W), consists mostly of well-bedded silts with minor sands and clays. The top of this unit is marked by a prominent disconformity (Disconformity A), and the base is marked by an unconformity. The underlying unit 2 contains the Little Timber Tephra which Westgate (Morlan, private communication) has dated at 1.2 Ma BP using fission track methods. Morlan (1986) deduces an age of about 39 000 yr for Disconformity A. Within unit 3 itself there occurs a widespread volcanic ash known as the Old Crow Tephra from which Westgate & Wintle (1986) have recently obtained a thermoluminescence date of 86 000 ± 8000 yr BP.

The samples were obtained by tapping small (8 cc) plastic boxes into prepared surfaces and orientation was achieved using a Brunton compass and a specially designed inclinometer. Twenty-six solar bearings determined the local magnetic declination to be 33°30′ ± 05′E. Two samples were collected from each stratigraphic horizon, the horizons being separated by 6 cm on average. In order to check the reality of any geomagnetic signal eventually obtained it was decided to sample two parallel sections (A, 196 samples, and B, 158 samples) separated laterally by 20 m. In both cases sampling began 0.5 m above the Old Crow Tephra and ended 5.5 m below it. Many of the individual stratigraphic sub-units, including the Tephra itself, could be traced directly from A to B, and this greatly facilitated correlation.

After removal from the outcrop, the samples were sealed to prevent loss of moisture, and on return to the laboratory they were stored in a magnetically shielded room in which the field is less than 50 nT. Remanent magnetizations were measured on a Digico balanced fluxgate spinner magnetometer, and alternating-field (AF) demagnetization was carried out using an apparatus similar to that described by McElhinny (1966).

C Results

All samples were subjected to partial AF demagnetization to eliminate unwanted viscous components of magnetization. To determine the appropriate treatment, ten samples were step-wise demagnetized up to peak fields of 100 mT. Up to 20 mT, movements were small but beyond this level some samples exhibited large erratic jumps. The average median destructive field was 19 ± 1 mT. On the basis of these observations it was decided to

demagnetize all remaining samples at 10 and 20 mT. Between 0 and 10 mT there was a general movement away from the present field at the sampling locality ($D = 033°, I = +82°$) (0 mT, $\bar{D} = 008°$, $\bar{I} = +72°$, $\alpha_{95} = 1.3°$; 10 mT, $\bar{D} = 351°$, $\bar{I} = +71°$, $\alpha_{95} = 1.6°$), but between 10 and 20 mT ($\bar{D} = 352°$, $\bar{I} = +71°$, $\alpha_{95} = 1.8°$) movement is negligible indicating that secondary components have been largely eliminated. We therefore concentrate further discussion on the 10 mT data.

Even after AF treatment, the two samples at certain horizons did not agree with one another. Since we were hoping ultimately to resolve secular variation features with magnitudes as small as 10°, we decided to simply eliminate any horizon at which the angular difference between the two measured vectors exceeded 10°. By some standards, this is a severe rejection criterion (e.g. the usual 5 per cent confidence level (R_0) for $N = 2$ is 1.949 which corresponds to an angular separation of 26°) and leads to the rejection of 38 horizons out of a total of 165. However, the main features of the resulting magnetograms are not affected and we therefore prefer to retain only the highest quality data. Prior to further analysis the two sections were combined on the basis of obvious geological correlations. Fig. 1 illustrates the resulting declination and inclination patterns.

D Discussion

Figure 2 shows the data of Fig. 1 after smoothing by a five-horizon (ten-sample) moving

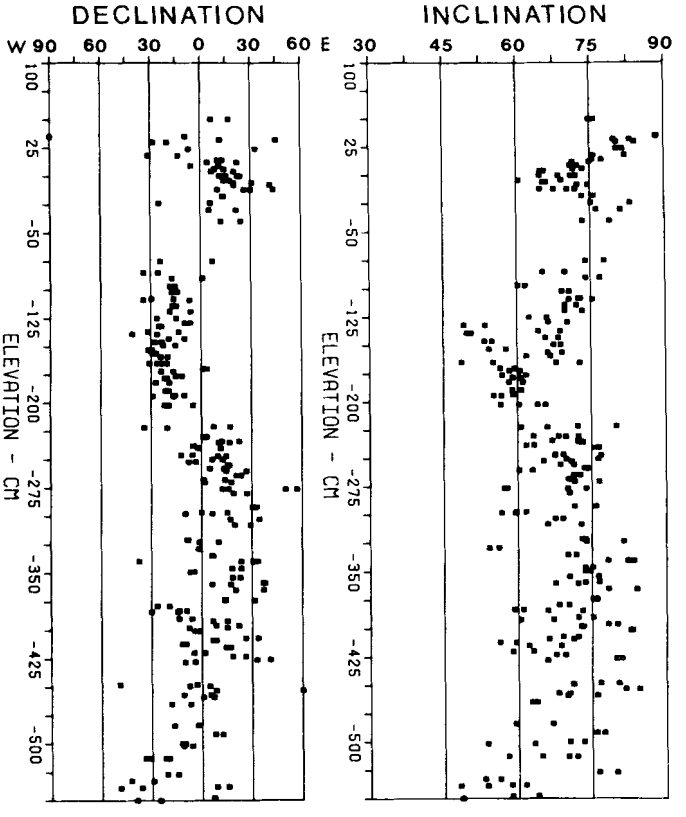


Figure 1. Declination and inclination magnetograms for sections A and B combined, after AF treatment at 10 mT. Elevations are relative to the Old Crow Tephra.

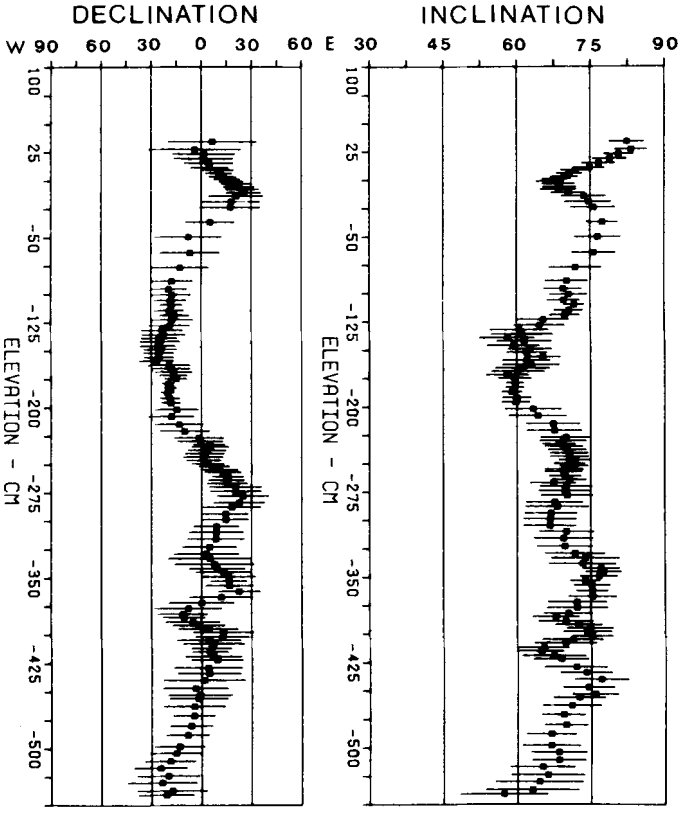


Figure 2. Smoothed declination and inclination magnetograms using a five-horizon (ten-sample) moving vector average. Error bars represent the Fisherian α_{95}.

vector average. Several more or less distinct topics emerge and it is convenient to discuss these separately in the following sub-sections.

ABSENCE OF THE BLAKE EVENT

One of the original goals of this project was to locate and investigate the Blake Event, which has been proposed as a global period of reversed polarity that occurred roughly 100 000 yr BP (Smith & Foster 1969; Sasajima, Nishimura & Hirooka 1980; Creer, Readman & Jacobs 1980). Indeed, Westgate *et al.* (1985) have reported reversed polarities some 2 m below the Old Crow Tephra in a locality only 35 km southwest of the present study area. Nevertheless, no convincing reversals were found in our sections A or B. The only relevant evidence is provided by a single pilot sample (A071; 188 cm below the Old Crow Tephra) whose remanence vector moves systematically towards the horizontal as AF demagnetization proceeds (Fig. 3). Other samples near this horizon were subjected to extended AF treatments but they did not exhibit the same type of behaviour. Thus, although sample A071 may carry an underlying reversed magnetization, the evidence is far from compelling. It appears, therefore, that the Blake Event is missing from our sections. The likely sedimentation rates (discussed below) suggest that our sampling interval is far too short to accommodate the entire Blake Event, and it therefore seems likely that a hiatus of some kind is responsible.

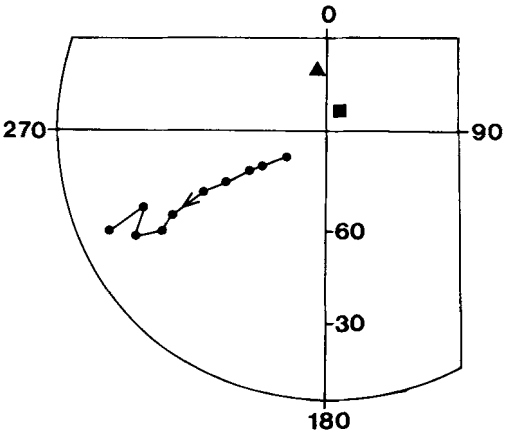


Figure 3. Behaviour during AF demagnetization of sample A071. Treatments are 0, 5, 10, 20, 30, 40, 50, 60, 70, and 80 mT. The square represents the present field at the sampling locality and the triangle the mean direction for the entire collection (at 10 mT). Equal-area projection, lower hemisphere.

STANDING SECULAR VARIATION SOURCES

It is instructive to combine the observed declination and inclination patterns on an equidistant azimuthal (Bauer) plot as in Fig. 4. This immediately reveals the type of pattern which can be attributed to a standing source. Westward (or eastward) drifting sources generally produce symmetrical secular variation (SV) in the form of open loops, whereas a stationary source which suddenly grows and decays leads to a linear perturbation involving an 'outward' and a superimposed 'return' trajectory (e.g. see Creer & Tucholka 1983). In the present case one clear example of this kind occurs between −241 cm and −11 cm (Fig. 4b), and part of a very similar signal is present in the lowermost 131 cm of the record (Fig. 4a). Evans (1984) has discussed other published examples of this type of behaviour, one of which involves a sedimentary sequence in British Columbia (Turner, Evans & Hussin 1982). This latter site lies some 18° (2000 km) to the south of our Old Crow site and although the ages

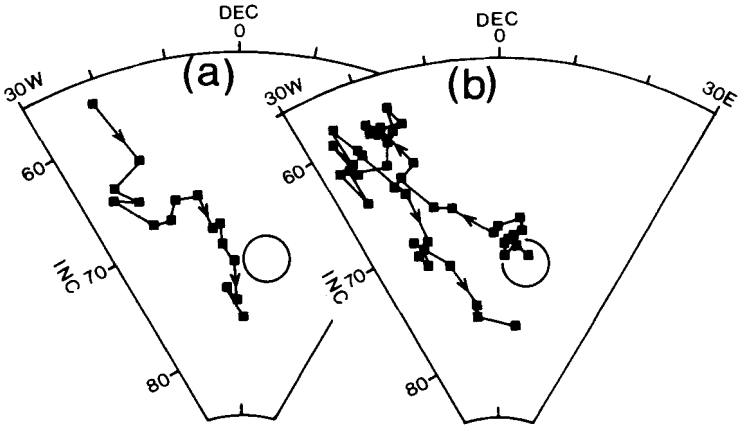


Figure 4. Linear secular variation perturbations in the intervals −550 to −419 cm (a), and −241 to −11 cm (b). The circle represents the α_{95} surrounding the mean direction obtained from the remaining 'unperturbed' intervals.

of the sediments are quite different – about 100 000 yr at Old Crow, about 20 000–30 000 yr in British Columbia – it seems worthwhile to enquire if the same source, acting intermittently, could account for both datasets. In this context we note that Creer & Tucholka (1982) have argued for the persistence of stationary sources over intervals of several thousands of years.

We start by considering a simple model in which the observed linear perturbations are caused by a stationary radial dipole located just below the core–mantle boundary (CMB). The perturbed and unperturbed vectors define a plane in which the perturbing vector itself must also lie. This constrains any source of magnetic flux at the CMB to lie on a curve. The curves corresponding to the perturbations observed in this study (between −241 and −11 cm) and by Turner *et al.* are illustrated in Fig. 5 (curves A and B respectively). For geometrical reasons each curve contains a gap within which it is impossible for a radial dipole to cause the observed perturbation. As can be seen in Fig. 5, the geometry is such that the curves corresponding to the two perturbations do not actually intersect one another, which means that, strictly speaking, the two datasets cannot be precisely modelled by means of a single source. However, if one allows the possibility that such a source may not have been perfectly radial and/or the possibility that some long-term migration may have occurred between the times of deposition of the sedimentary sequences involved, both perturbations could have had a common source. At the point of maximum perturbation, the geomagnetic field vector in this study lies 22° away from its unperturbed orientation, whereas the perturbation observed by Turner *et al.* involves 20° of movement. Using elementary magnetostatic theory it is easy to determine how the required magnetic moment

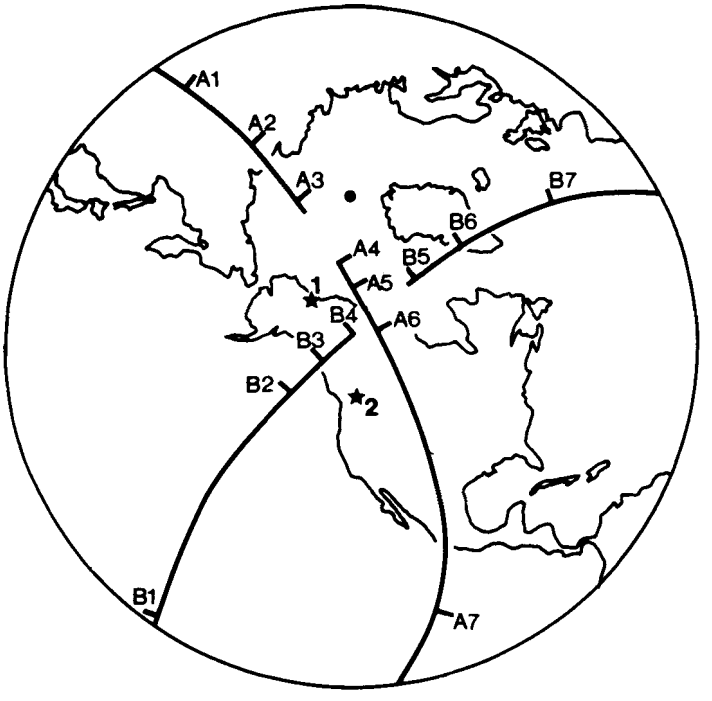


Figure 5. Map showing the location of the present study area (1), and that of Turner *et al.* (1982) (2). The curves show the loci of radial dipoles at the core–mantle boundary capable of explaining the extreme perturbation seen at site 1 (curve A) and at site 2 (curve B). The dot indicates the North Pole (see text and Fig. 6 for further explanation).

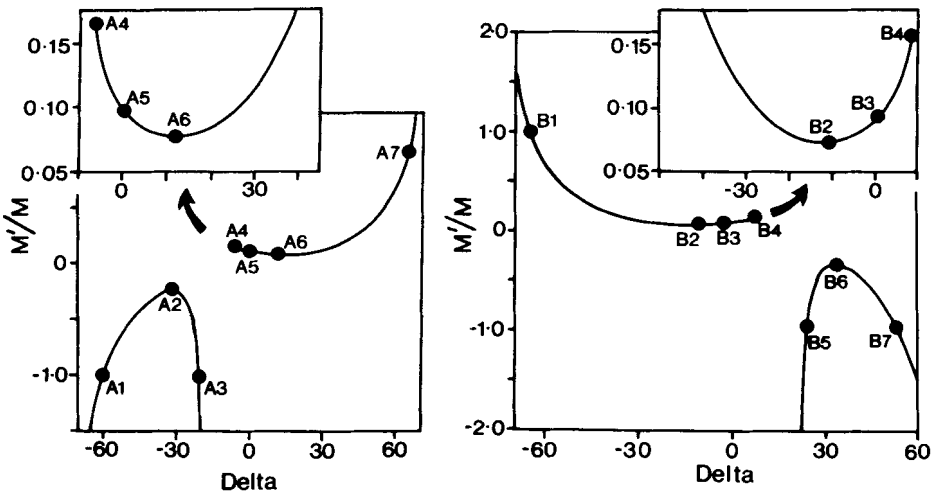


Figure 6. Graphs illustrating the magnetic moment of a radial dipole necessary to explain the extreme perturbations observed. The points are labelled as in Fig. 5. Points A5 and B3 are the arbitrary origins, being the points of closest approach to sites 1 and 2 respectively. M′ is the moment of the radial dipole in question, M is the moment of the geocentric dipole (see text for further explanation).

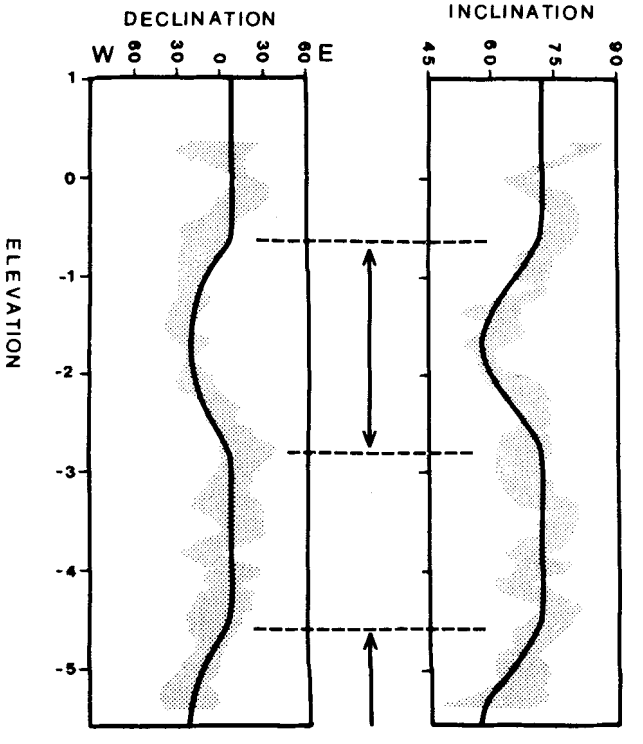


Figure 7. Synthetic magnetograms (heavy lines) compared to the observations (shaded areas correspond to the error bars in Fig. 2). The arrows indicate when the subsidiary dipoles are active, during which time they vary as a half sine-wave.

(M′) of the radial dipole varies along curves A and B (Fig. 6). In both cases, the minimum value of M′ lies between 7 per cent and 8 per cent of the Earth's main dipole moment (M). Although it is not known how strong such dipoles may become, it is perhaps reasonable to accept $M'/M = \pm 1.0$ as an upper limit. This restricts any candidate source to the 75° between points A4 and A7, or the 49° between A1 and A3. Comparable results are found for the perturbation observed by Turner *et al.* The minima in the two curves (points A6 and B2, Fig. 5) lies only 15° apart, thus the migratory motion required for a radial dipole of this magnitude to cause both perturbations is not great.

As an illustration of the effectiveness of this simple model, Fig. 7 shows the magnetograms obtained with a dipole situated at point A6 which is allowed to vary sinusoidally in time from zero up to a maximum of $M'/M = 0.08$.

FAR-SIDEDNESS OF THE AVERAGE VGP

The average virtual geomagnetic pole (VGP) from the section sampled in this study does not coincide with the geographic axis but lies at 75.6°N, 49.4°E ($\alpha_{95} = 2.7°$). Using Wilson's terminology (Wilson 1971) this is both far-sided and left-handed. The perturbations discussed above involve relatively low inclinations so some of the far-sidedness is inevitably due to them. Considering only those points that do not lie in the perturbed intervals (between −241 and −11 cm and below −419 cm) the average VGP is still far-sided (by 11°) and lies at 79.1°N, 19.4°E ($\alpha_{95} = 3.3°$). Such far-sidedness could result from a number of distinct causes, the most important of which are (a) landslip, (b) depositional inclination error, (c) a permanent quadrupole moment, and (d) dipole wobble.

(a) In addition to simple downward translation, landslip can also lead to a rigid-body rotation. However the strata sampled are horizontal and there is no visual evidence of landslip. Furthermore, the orientation of the river valley at this site is such that declination might be affected but little, or no, deviation in inclination can be expected.

(b) Several authors (Johnson, Murphy & Torreson 1948; King 1955; King & Rees 1966; Blow & Hamilton 1978) have shown that artificially deposited sediments often preserve an inclination significantly shallower than that of the ambient field. Discrepancies in excess of 20° are quite possible and thus the relatively small polar offset found in the present study (14° for the complete dataset, 11° if the perturbations are excluded) could well be due to such a sedimentological cause. However there is a growing body of evidence (Barton & McElhinny 1979; Tucker 1983) that real sediments actually carry a post-depositional remanence (PDRM) which enables grain re-adjustment in water-filled voids such that the inclination discrepancy is greatly reduced or even entirely removed. We have not carried out laboratory re-deposition experiments on the Old Crow sediments, but there is evidence that they were capable of fluid-assisted rotations some time after deposition. This is provided by a comparison between results from a thin cryoturbated zone with those from a laterally equivalent undisturbed zone. Eight samples were collected between depths of −218 and −250 cm in section A and a further eight were taken from an equivalent cryoturbated zone a few metres to the side of section A itself. The disturbance of the sediments is quite marked, but the internal precision is the same in the two groups (Table 1). Initially, this was rather surprising, but if one assumes that cryoturbation involves freezing and thawing with the intermittent presence of interstitial water, then a later re-setting of the remanence is quite plausible. As well as the similar precisions of the two groups, it was found that the inclinations and intensities were not significantly different (Table 1); but the declinations do differ, being 18 ± 10° in the undisturbed sediments and 353 ± 12° in the cryoturbated zone. Inspection of the complete dataset indicates that this declination could refer to the time of

Table 1. Comparison of results from cryoturbated and undisturbed sediments

	N	*D*	*I*	*M*
Cryoturbated	8	353 ± 12	72 ± 4	1.3 ± 0.1
Undisturbed	8	018 ± 10	73 ± 3	1.2 ± 0.1

N = number of samples; D = mean declination in degrees E of N (uncertainty is $\alpha_{95}/\cos I$); I = mean inclination in degrees below horizontal (uncertainty is α_{95}); M = mean magnetic intensity in Am^{-1} (uncertainty is standard error).

deposition some 75 cm higher in the sequence. The exact time is, of course, unimportant in the present context, what is important is the preservation of a well-defined remanence despite the marked cryoturbation. This demonstrates that the internal structure of these sediments is such that localized grain rotations are possible which, in turn, implies that the water present during and shortly after deposition would very probably have permitted an 'original' PDRM.

(c) A permanent quadrupole moment superimposed on the main central dipole can produce global far-sidedness (Wilson 1971), but at high latitudes very large moments are required to cause more than a few degrees of offset. To explain the observed discrepancy of 11°, the required axial quadrupole moment exceeds 60 per cent of the main dipole moment ($g_2^0/g_1^0 = 0.61$). This is ten times larger than the present value and it seems improbable, therefore, that the observed far-sidedness could be entirely due to a quadrupole component.

(d) Finally, then, it appears that the 6 m of sediments sampled in this study comprise only a fraction of the time required to average out the effect of dipole wobble. Support for this contention is provided by the probable sedimentation rates and by the observed dispersion of the virtual geomagnetic poles (VGPs). Kukal (1971) has comprehensively reviewed the rates of sedimentation to be expected in a variety of environments, and although there is considerable variation, it seems likely that the sediments were deposited at an average rate between 10 and 100 cm per 1000 years. This corresponds to a total time interval between 6×10^4 and 6×10^3 yr, which is probably insufficient to fully sample the dipole wobble. The angular dispersion of the VGP's is 16.3 ± 1.4° (N=127) which is about 4° lower than that predicted by Model F (McFadden & McElhinny 1984) for this latitude, which again favours a relatively short time-interval. If one removes the best-fit dipole wobble component of the dispersion obtained by McFadden & McElhinny, one is left with a dispersion of 18° arising from non-dipole sources. This lies just outside the limits of uncertainty obtained from the Old Crow data and hence it seems reasonable to suppose that the observed scatter is due to non-dipole variations with little or no dipole wobble contribution. This, in turn, supports the contention that the observed far-sidedness reflects nothing more than a 'snapshot' of dipole wobble.

E Conclusions

One of the original goals of this work was to locate and investigate the Blake Event, partly because of its use as a stratigraphic marker and partly for its own sake as a genuine short-lived global polarity reversal. Apart from the tenuous evidence of a single sample whose behaviour during AF demagnetization suggests the presence of an underlying reversed magnetization, no trace of the Blake Event was found. It appears that a hiatus due either to non-deposition or to erosion has prevented the recording and/or the recovery of the

appropriate data. However, the secular variation pattern revealed is interesting because it provides further evidence for the existence in the geological past of stationary magnetic sources in the core. The observed pattern takes the form of a spatially isotropic dispersion due to typical non-dipole field behaviour as described by the model of McFadden & McElhinny (1984), on which is superimposed linear perturbations due to stationary sources as described by Evans (1984). These linear perturbations – which are characterized by coincident 'outward' and 'return' trajectories – are readily modelled by elementary magnetostatics. In the present study it is found that a radial dipole situated just below the core–mantle boundary and having a magnetic moment as low as 8 per cent of that of the main central dipole is sufficient to explain the maximum observed perturbation. Subsidiary dipoles of this kind are reminiscent of Cox's (1975) model of spatially non-isotropic secular variation and may correspond to the 'static flux bundles' recently recognized by Bloxham & Gubbins (1985) in their novel analysis of the historical geomagnetic records. The intriguing possibility of very long-lived stationary sources in the outer core is therefore raised. Furthermore, if they are biased in such a way that more of them point outwards than inwards, then the well-known persistent quadrupole component of the geomagnetic field is readily explained.

Acknowledgments

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